

PHP a MySQL – MySQLi, PDO

Staré řešení

```
<html><body>
<table border=1 cellpadding=2>
<tr>
    <th>ID</th>
    <th>Firma</th>
    <th>Kontakt</th>
</tr>

<?php
$id=$_GET["id"];

$mysql = MySQL_Connect("localhost","smrcka","dbsmrcka");
$spojeni=MySQL_Select_DB("smrcka");

$sql="SELECT id_zak, firma, kontakt_jmeno FROM zakaznici WHERE id_zak = ".$id."";
$vys = mysql_query($sql);

while($zaznam=mysql_Fetch_array($vys))
{
    echo "<tr>";
    echo "<td>".$zaznam["id_zak"]."</td>";
    echo "<td>".$zaznam["firma"]."</td>";
    echo "<td>".$zaznam["kontakt_jmeno"]."</td>";
    echo "</tr>";
}
echo "</table>";
mysql_close();
?>
</body>
</html>
```

PHP Connect to MySQL

Zdroj https://www.w3schools.com/php/php_mysql_connect.asp

Example (MySQLi Object-Oriented)

```
<?php
$servername = "localhost";
$username = "username";
$password = "password";

// Create connection
$conn = new mysqli($servername, $username, $password);
```

```
// Check connection
if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}
echo "Connected successfully";
?>
```

Example (MySQLi Procedural)

```
<?php
$servername = "localhost";
$username = "username";
$password = "password";

// Create connection
$conn = mysqli_connect($servername, $username, $password);

// Check connection
if (!$conn) {
    die("Connection failed: " . mysqli_connect_error());
}
echo "Connected successfully";
?>
```

Example (PDO)

```
<?php
$servername = "localhost";
$username = "username";
$password = "password";

try {
    $conn = new PDO("mysql:host=$servername;dbname=myDB", $username,
$password);

    $this->conn->exec("set names utf8");
    // set the PDO error mode to exception
    $conn->setAttribute(PDO::ATTR_ERRMODE, PDO::ERRMODE_EXCEPTION);
    echo "Connected successfully";
} catch(PDOException $e) {
    echo "Connection failed: " . $e->getMessage();
}
```

```
}  
?>
```

Příklad řešení mysqli

```
<html><body>  
<table border=1 cellpadding=2>  
<tr>  
    <th>ID</th>  
    <th>Firma</th>  
    <th>Kontakt</th>  
</tr>  
  
<?php  
$id=$_GET["id"];  
  
$mysqli = new mysqli('localhost', 'smrcka', 'dbsmrcka', 'smrcka');  
  
if ($mysqli->connect_error) {  
    die('Nepodarilo se pripojit k MySQL serveru (' . $mysqli->connect_errno . ') '  
        . $mysqli->connect_error);  
}  
  
$vysledek = $mysqli->query("SELECT * FROM zakaznici where id_zak = '".$id."' ");  
  
while ($uzivatel = $vysledek->fetch_assoc())  
{  
    echo "<tr>";  
    echo "<td>".$uzivatel['id']. "</td>";  
    echo "<td>".$uzivatel['jmeno']. "</td>";  
    echo "<td>".$uzivatel['firma']. "</td>";  
  
    echo "<td>".$uzivatel['kontakt_jmeno']. "</td>";  
    echo "</tr>";  
}  
  
echo "</table>";  
$mysqli->close();  
  
?>  
</body>  
</html>
```

Insert

Zdroj

https://www.w3schools.com/php/php_mysql_insert.asp

Example (MySQLi Object-oriented)

```
<?php
$servername = "localhost";
$username = "username";
$password = "password";
$dbname = "myDB";

// Create connection
$conn = new mysqli($servername, $username, $password, $dbname);
// Check connection
if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}

$sql = "INSERT INTO MyGuests (firstname, lastname, email)
VALUES ('John', 'Doe', 'john@example.com')";

if ($conn->query($sql) === TRUE) {
    echo "New record created successfully";
} else {
    echo "Error: " . $sql . "<br>" . $conn->error;
}

$conn->close();
?>
```

ADVERTISEMENT

Example (MySQLi Procedural)

```
<?php
$servername = "localhost";
$username = "username";
$password = "password";
$dbname = "myDB";

// Create connection
$conn = mysqli_connect($servername, $username, $password, $dbname);
// Check connection
if (!$conn) {
    die("Connection failed: " . mysqli_connect_error());
}

$sql = "INSERT INTO MyGuests (firstname, lastname, email)
VALUES ('John', 'Doe', 'john@example.com')";

if (mysqli_query($conn, $sql)) {
    echo "New record created successfully";
} else {
    echo "Error: " . $sql . "<br>" . mysqli_error($conn);
}

mysqli_close($conn);
?>
```

Example (PDO)

```
<?php
$servername = "localhost";
$username = "username";
$password = "password";
$dbname = "myDBPDO";

try {
    $conn = new PDO("mysql:host=$servername;dbname=$dbname", $username,
$password);
    // set the PDO error mode to exception
    $conn->setAttribute(PDO::ATTR_ERRMODE, PDO::ERRMODE_EXCEPTION);
    $sql = "INSERT INTO MyGuests (firstname, lastname, email)
VALUES ('John', 'Doe', 'john@example.com')";
```

```
// use exec() because no results are returned
$conn->exec($sql);
echo "New record created successfully";
} catch(PDOException $e) {
    echo $sql . "<br>" . $e->getMessage();
}
```

\$conn = null;

?>

Get last ID

[https://www.w3schools.com/php/
php_mysql_insert_lastid.asp](https://www.w3schools.com/php/php_mysql_insert_lastid.asp)

Insert Multiple Records

[https://www.w3schools.com/php/
php_mysql_insert_multiple.asp](https://www.w3schools.com/php/php_mysql_insert_multiple.asp)

Example (MySQLi Object-oriented)

<?php

\$servername = "localhost";

\$username = "username";

\$password = "password";

\$dbname = "myDB";

// Create connection

\$conn = new mysqli(\$servername, \$username, \$password, \$dbname);

// Check connection

if (\$conn->connect_error) {

die("Connection failed: " . \$conn->connect_error);

}

\$sql = "INSERT INTO MyGuests (firstname, lastname, email)

```
VALUES ('John', 'Doe', 'john@example.com');";
$sql .= "INSERT INTO MyGuests (firstname, lastname, email)
VALUES ('Mary', 'Moe', 'mary@example.com');";
$sql .= "INSERT INTO MyGuests (firstname, lastname, email)
VALUES ('Julie', 'Dooley', 'julie@example.com')";
```

```
if ($conn->multi_query($sql) === TRUE) {
    echo "New records created successfully";
} else {
    echo "Error: " . $sql . "<br>" . $conn->error;
}
```

```
$conn->close();
?>
```

Note that each SQL statement must be separated by a semicolon.

Example (MySQLi Procedural)

```
<?php
$servername = "localhost";
$username = "username";
$password = "password";
$dbname = "myDB";

// Create connection
$conn = mysqli_connect($servername, $username, $password, $dbname);
// Check connection
if (!$conn) {
    die("Connection failed: " . mysqli_connect_error());
}

$sql = "INSERT INTO MyGuests (firstname, lastname, email)
VALUES ('John', 'Doe', 'john@example.com');";
$sql .= "INSERT INTO MyGuests (firstname, lastname, email)
VALUES ('Mary', 'Moe', 'mary@example.com');";
$sql .= "INSERT INTO MyGuests (firstname, lastname, email)
VALUES ('Julie', 'Dooley', 'julie@example.com')";

if (mysqli_multi_query($conn, $sql)) {
    echo "New records created successfully";
```

```

} else {
    echo "Error: " . $sql . "<br>" . mysqli_error($conn);
}

```

```

mysqli_close($conn);

```

```

?>

```

The PDO way is a little bit different:

Example (PDO)

```

<?php

```

```

$servername = "localhost";

```

```

$username = "username";

```

```

$password = "password";

```

```

$dbname = "myDBPDO";

```

```

try {

```

```

    $conn = new PDO("mysql:host=$servername;dbname=$dbname", $username,
$password);

```

```

    // set the PDO error mode to exception

```

```

    $conn->setAttribute(PDO::ATTR_ERRMODE, PDO::ERRMODE_EXCEPTION);

```

```

    // begin the transaction

```

```

    $conn->beginTransaction();

```

```

    // our SQL statements

```

```

    $conn->exec("INSERT INTO MyGuests (firstname, lastname, email)
VALUES ('John', 'Doe', 'john@example.com')");

```

```

    $conn->exec("INSERT INTO MyGuests (firstname, lastname, email)
VALUES ('Mary', 'Moe', 'mary@example.com')");

```

```

    $conn->exec("INSERT INTO MyGuests (firstname, lastname, email)
VALUES ('Julie', 'Dooley', 'julie@example.com')");

```

```

    // commit the transaction

```

```

    $conn->commit();

```

```

    echo "New records created successfully";

```

```

} catch(PDOException $e) {

```

```

    // roll back the transaction if something failed

```

```

    $conn->rollback();

```

```

    echo "Error: " . $e->getMessage();

```



```
}
```

```
$conn = null;
```

```
?>
```

PHP MySQL Prepared Statements

Připravené příkazy jsou velmi užitečné proti SQL injekcím.

[https://www.w3schools.com/php/
php_mysql_prepared_statements.asp](https://www.w3schools.com/php/php_mysql_prepared_statements.asp)

Připravená prohlášení v zásadě fungují takto:

1. Příprava: Vytvoří se šablona příkazu SQL a odešle se do databáze. Určité hodnoty zůstávají nespecifikované, nazývají se parametry (označené „?“). Příklad: INSERT INTO MyGuests VALUES (?, ?, ?)
2. Databáze analyzuje, kompiluje a provádí optimalizaci dotazu na šabloně příkazu SQL a ukládá výsledek bez jeho provedení
3. Provést: Později aplikace váže hodnoty na parametry a databáze provede příkaz. Aplikace může provést příkaz tolikrát, kolikrát chce, s různými hodnotami

Následující příklad používá připravené příkazy a vázané parametry v MySQLi:

Příklad (MySQLi s připravenými příkazy)

```
<?php
```

```
$servername = "localhost";
```

```
$username = "username";
```

```
$password = "password";
```

```
$dbname = "myDB";
```

```
// Create connection
```

```
$conn = new mysqli($servername, $username, $password, $dbname);
```

```
// Check connection
```

```
if ($conn->connect_error) {
```

```
    die("Connection failed: " . $conn->connect_error);
```

```
}
```

```
// prepare and bind
```

```
$stmt = $conn->prepare("INSERT INTO MyGuests (firstname, lastname, email)
VALUES (?, ?, ?)");
$stmt->bind_param("sss", $firstname, $lastname, $email);
```

```
// set parameters and execute
```

```
$firstname = "John";
$lastname = "Doe";
$email = "john@example.com";
$stmt->execute();
```

```
$firstname = "Mary";
$lastname = "Moe";
$email = "mary@example.com";
$stmt->execute();
```

```
$firstname = "Julie";
$lastname = "Dooley";
$email = "julie@example.com";
$stmt->execute();
```

```
echo "New records created successfully";
```

```
$stmt->close();
$conn->close();
?>
```

Řádky kódu k vysvětlení z výše uvedeného příkladu:

```
"INSERT INTO MyGuests (firstname, lastname, email) VALUES (?, ?, ?)"
```

V našem SQL vložíme otazník (?), Kde chceme nahradit celou hodnotu, řetězec, dvojitou nebo blobovou hodnotu.

Pak se podívejte na funkci bind_param ():

```
$stmt->bind_param("sss", $firstname, $lastname, $email);
```

Tato funkce váže parametry na dotaz SQL a říká databázi, jaké jsou parametry. Argument "sss" uvádí seznam typů dat, kterými jsou parametry. Znak s říká mysql, že parametr je řetězec.

Argument může být jeden ze čtyř typů:

- i - celé číslo
- d - dvojnásobek
- s - řetězec
- b - BLOB

Pro každý parametr musíme mít jednu z nich.

Tím, že řekneme mysql, jaký typ dat očekávat, minimalizujeme riziko injekcí SQL.

Poznámka: Pokud chceme vložit libovolná data z externích zdrojů (například vstup uživatele), je velmi důležité, aby byla data dezinfikována a ověřena.

Example (PDO with Prepared Statements)

```
<?php
```

```
$servername = "localhost";
```

```
$username = "username";
```

```
$password = "password";
```

```
$dbname = "myDBPDO";
```

```
try {
```

```
    $conn = new PDO("mysql:host=$servername;dbname=$dbname", $username,
    $password);
```

```
    // set the PDO error mode to exception
```

```
    $conn->setAttribute(PDO::ATTR_ERRMODE, PDO::ERRMODE_EXCEPTION);
```

```
    // prepare sql and bind parameters
```

```
    $stmt = $conn->prepare("INSERT INTO MyGuests (firstname, lastname, email)
    VALUES (:firstname, :lastname, :email)");
```

```
    $stmt->bindParam(':firstname', $firstname);
```

```
    $stmt->bindParam(':lastname', $lastname);
```

```
    $stmt->bindParam(':email', $email);
```

```
    // insert a row
```

```
    $firstname = "John";
```

```
    $lastname = "Doe";
```

```
    $email = "john@example.com";
```

```
    $stmt->execute();
```

```
    // insert another row
```

```
    $firstname = "Mary";
```

```
    $lastname = "Moe";
```

```
    $email = "mary@example.com";
```

```

$stmt->execute();

// insert another row
$firstname = "Julie";
$lastname = "Dooley";
$email = "julie@example.com";
$stmt->execute();

echo "New records created successfully";
} catch(PDOException $e) {
    echo "Error: " . $e->getMessage();
}
$conn = null;
?>

```

Select Data From a MySQL Database

https://www.w3schools.com/php/php_mysql_select.asp

Example (MySQLi Object-oriented)

```

<?php
$servername = "localhost";
$username = "username";
$password = "password";
$dbname = "myDB";

// Create connection
$conn = new mysqli($servername, $username, $password, $dbname);
// Check connection
if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}

$sql = "SELECT id, firstname, lastname FROM MyGuests";
$result = $conn->query($sql);

if ($result->num_rows > 0) {
    // output data of each row
    while($row = $result->fetch_assoc()) {

```

```

        echo "id: " . $row["id"]. " - Name: " . $row["firstname"]. " " .
$row["lastname"]. "<br>";
    }
} else {
    echo "0 results";
}
$conn->close();
?>

```

Example (MySQLi Procedural)

```

<?php
$servername = "localhost";
$username = "username";
$password = "password";
$dbname = "myDB";

// Create connection
$conn = mysqli_connect($servername, $username, $password, $dbname);
// Check connection
if (!$conn) {
    die("Connection failed: " . mysqli_connect_error());
}

$sql = "SELECT id, firstname, lastname FROM MyGuests";
$result = mysqli_query($conn, $sql);

if (mysqli_num_rows($result) > 0) {
    // output data of each row
    while($row = mysqli_fetch_assoc($result)) {
        echo "id: " . $row["id"]. " - Name: " . $row["firstname"]. " " .
$row["lastname"]. "<br>";
    }
} else {
    echo "0 results";
}

mysqli_close($conn);
?>

```

You can also put the result in an HTML table:

Example (MySQLi Object-oriented)

```
<?php
$servername = "localhost";
$username = "username";
$password = "password";
$dbname = "myDB";

// Create connection
$conn = new mysqli($servername, $username, $password, $dbname);
// Check connection
if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}

$sql = "SELECT id, firstname, lastname FROM MyGuests";
$result = $conn->query($sql);

if ($result->num_rows > 0) {
    echo "<table><tr><th>ID</th><th>Name</th></tr>";
    // output data of each row
    while($row = $result->fetch_assoc()) {
        echo "<tr><td>".$row["id"]."</td><td>".$row["firstname"]." ".
        $row["lastname"]."</td></tr>";
    }
    echo "</table>";
} else {
    echo "0 results";
}
$conn->close();
?>
```

Select and Filter Data From a MySQL Database

https://www.w3schools.com/php/php_mysql_select_where.asp

Example (MySQLi Object-oriented)

```
<?php
$servername = "localhost";
$username = "username";
$password = "password";
```

```

$dbname = "myDB";

// Create connection
$conn = new mysqli($servername, $username, $password, $dbname);
// Check connection
if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}

$sql = "SELECT id, firstname, lastname FROM MyGuests WHERE lastname='Doe'";
$result = $conn->query($sql);

if ($result->num_rows > 0) {
    // output data of each row
    while($row = $result->fetch_assoc()) {
        echo "id: " . $row["id"]. " - Name: " . $row["firstname"]. " " .
        $row["lastname"]. "<br>";
    }
} else {
    echo "0 results";
}
$conn->close();
?>

```

Example (MySQLi Procedural)

```

<?php
$servername = "localhost";
$username = "username";
$password = "password";
$dbname = "myDB";

// Create connection
$conn = mysqli_connect($servername, $username, $password, $dbname);
// Check connection
if (!$conn) {
    die("Connection failed: " . mysqli_connect_error());
}

$sql = "SELECT id, firstname, lastname FROM MyGuests WHERE lastname='Doe'";
$result = mysqli_query($conn, $sql);

```

```

if (mysqli_num_rows($result) > 0) {
    // output data of each row
    while($row = mysqli_fetch_assoc($result)) {
        echo "id: " . $row["id"]. " - Name: " . $row["firstname"]. " " .
        $row["lastname"]. "<br>";
    }
} else {
    echo "0 results";
}

mysqli_close($conn);
?>

```

Example (MySQLi Object-oriented)

```

<?php
$servername = "localhost";
$username = "username";
$password = "password";
$dbname = "myDB";

// Create connection
$conn = new mysqli($servername, $username, $password, $dbname);
// Check connection
if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}

$sql = "SELECT id, firstname, lastname FROM MyGuests WHERE lastname='Doe'";
$result = $conn->query($sql);

if ($result->num_rows > 0) {
    echo "<table><tr><th>ID</th><th>Name</th></tr>";
    // output data of each row
    while($row = $result->fetch_assoc()) {
        echo "<tr><td>". $row["id"]. "</td><td>". $row["firstname"]. " " .
        $row["lastname"]. "</td></tr>";
    }
    echo "</table>";
}

```



```

} else {
    echo "0 results";
}
$conn->close();
?>

```

Example (PDO)

```

<?php
echo "<table style='border: solid 1px black;'>";
echo "<tr><th>Id</th><th>Firstname</th><th>Lastname</th></tr>";

class TableRows extends RecursiveIteratorIterator {
    function __construct($it) {
        parent::__construct($it, self::LEAVES_ONLY);
    }

    function current() {
        return "<td style='width:150px;border:1px solid black;'>" .
parent::current(). "</td>";
    }

    function beginChildren() {
        echo "<tr>";
    }

    function endChildren() {
        echo "</tr>" . "\n";
    }
}

$servername = "localhost";
$username = "username";
$password = "password";
$dbname = "myDBPDO";

try {
    $conn = new PDO("mysql:host=$servername;dbname=$dbname", $username,
$password);
    $conn->setAttribute(PDO::ATTR_ERRMODE, PDO::ERRMODE_EXCEPTION);
    $stmt = $conn->prepare("SELECT id, firstname, lastname FROM MyGuests

```

```

WHERE lastname='Doe');
$stmt->execute();

// set the resulting array to associative
$result = $stmt->setFetchMode(PDO::FETCH_ASSOC);
foreach(new TableRows(new RecursiveArrayIterator($stmt->fetchAll())) as $k=>$
v) {
    echo $v;
}
}
catch(PDOException $e) {
    echo "Error: " . $e->getMessage();
}
$conn = null;
echo "</table>";
?>

```

PHP MySQL Delete Data

https://www.w3schools.com/php/php_mysql_delete.asp

Example (MySQLi Object-oriented)

```

<?php
$servername = "localhost";
$username = "username";
$password = "password";
$dbname = "myDB";

// Create connection
$conn = new mysqli($servername, $username, $password, $dbname);
// Check connection
if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}

// sql to delete a record
$sql = "DELETE FROM MyGuests WHERE id=3";

if ($conn->query($sql) === TRUE) {
    echo "Record deleted successfully";
}

```

```
} else {  
    echo "Error deleting record: " . $conn->error;  
}
```

```
$conn->close();
```

```
?>
```

Example (MySQLi Procedural)

```
<?php
```

```
$servername = "localhost";
```

```
$username = "username";
```

```
$password = "password";
```

```
$dbname = "myDB";
```

```
// Create connection
```

```
$conn = mysqli_connect($servername, $username, $password, $dbname);
```

```
// Check connection
```

```
if (!$conn) {
```

```
    die("Connection failed: " . mysqli_connect_error());
```

```
}
```

```
// sql to delete a record
```

```
$sql = "DELETE FROM MyGuests WHERE id=3";
```

```
if (mysqli_query($conn, $sql)) {
```

```
    echo "Record deleted successfully";
```

```
} else {
```

```
    echo "Error deleting record: " . mysqli_error($conn);
```

```
}
```

```
mysqli_close($conn);
```

```
?>
```

Example (PDO)

```
<?php
```

```
$servername = "localhost";
```

```
$username = "username";
```

```
$password = "password";
```

```
$dbname = "myDBPDO";
```

```

try {
    $conn = new PDO("mysql:host=$servername;dbname=$dbname", $username,
$password);
    // set the PDO error mode to exception
    $conn->setAttribute(PDO::ATTR_ERRMODE, PDO::ERRMODE_EXCEPTION);

    // sql to delete a record
    $sql = "DELETE FROM MyGuests WHERE id=3";

    // use exec() because no results are returned
    $conn->exec($sql);
    echo "Record deleted successfully";
} catch(PDOException $e) {
    echo $sql . "<br>" . $e->getMessage();
}

$conn = null;
?>

```

PHP MySQL Update Data

https://www.w3schools.com/php/php_mysql_update.asp

Example (MySQLi Object-oriented)

```

<?php
$servername = "localhost";
$username = "username";
$password = "password";
$dbname = "myDB";

// Create connection
$conn = new mysqli($servername, $username, $password, $dbname);
// Check connection
if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}

$sql = "UPDATE MyGuests SET lastname='Doe' WHERE id=2";

if ($conn->query($sql) === TRUE) {
    echo "Record updated successfully";
} else {

```

```
    echo "Error updating record: " . $conn->error;
}
```

```
$conn->close();
```

```
?>
```

Example (MySQLi Procedural)

```
<?php
```

```
$servername = "localhost";
```

```
$username = "username";
```

```
$password = "password";
```

```
$dbname = "myDB";
```

```
// Create connection
```

```
$conn = mysqli_connect($servername, $username, $password, $dbname);
```

```
// Check connection
```

```
if (!$conn) {
```

```
    die("Connection failed: " . mysqli_connect_error());
```

```
}
```

```
$sql = "UPDATE MyGuests SET lastname='Doe' WHERE id=2";
```

```
if (mysqli_query($conn, $sql)) {
```

```
    echo "Record updated successfully";
```

```
} else {
```

```
    echo "Error updating record: " . mysqli_error($conn);
```

```
}
```

```
mysqli_close($conn);
```

```
?>
```

Example (PDO)

```
<?php
```

```
$servername = "localhost";
```

```
$username = "username";
```

```
$password = "password";
```

```
$dbname = "myDBPDO";
```

```
try {
```

```
    $conn = new PDO("mysql:host=$servername;dbname=$dbname", $username,
```

```
$password);  
// set the PDO error mode to exception  
$conn->setAttribute(PDO::ATTR_ERRMODE, PDO::ERRMODE_EXCEPTION);  
  
$sql = "UPDATE MyGuests SET lastname='Doe' WHERE id=2";  
  
// Prepare statement  
$stmt = $conn->prepare($sql);  
  
// execute the query  
$stmt->execute();  
  
// echo a message to say the UPDATE succeeded  
echo $stmt->rowCount() . " records UPDATED successfully";  
} catch(PDOException $e) {  
    echo $sql . "<br>" . $e->getMessage();  
}  
  
$conn = null;  
?>
```